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# Logistic Regression - Complete Program

# Step 1: Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Step 2: Load the dataset
# Replace 'dataset.csv' with your dataset file name
data = pd.read_csv("dataset.csv")

# Step 3: Separate features (X) and target (y)
# Replace 'target' with the name of your target column
X = data.drop("target", axis=1)
y = data["target"]

# Step 4: Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Step 5: Initialize the Logistic Regression model
model = LogisticRegression(max_iter=1000)

# Step 6: Train the model
model.fit(X_train, y_train)

# Step 7: Predict on the test data
y_pred = model.predict(X_test)

# Step 8: Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", round(accuracy * 100, 2), "%")

# Step 9: Display confusion matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Step 10: Display classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Step 11: Compare actual and predicted values
comparison = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print("\nActual vs Predicted:")
print(comparison.head(10))
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FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_2591/611794722.py in <cell line: 0>()
      9 # Step 2: Load the dataset
     10 # Replace 'dataset.csv' with your dataset file name
----> 11 data = pd.read_csv("dataset.csv")
     12
     13 # Step 3: Separate features (X) and target (y)

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression,
memory_map, is_text, errors, storage_options)
     871     if ioargs.encoding and "b" not in ioargs.mode:
     872         # Encoding
--> 873         handle = open(
     874             handle,
     875             ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'dataset.csv'

```

Next steps: [Explain error](#)

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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset (Replace with your dataset name)
data = pd.read_csv("your_dataset.csv")

# Display first five rows
print(data.head())

# Replace 'target' with your target column name
X = data.drop("target", axis=1)
y = data["target"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=1000)

# Train the model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("\nAccuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

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FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_2591/1986379976.py in <cell line: 0>()
      5
      6 # Load dataset (Replace with your dataset name)
----> 7 data = pd.read_csv("your_dataset.csv")
      8
      9 # Display first five rows

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression,
memory_map, is_text, errors, storage_options)
      871     if ioargs.encoding and "b" not in ioargs.mode:
      872         # Encoding
--> 873         handle = open(
      874             handle,
      875             ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'your_dataset.csv'

```

Next steps: [Explain error](#)

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# Logistic Regression using a built-in dataset (No CSV file required)

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import pandas as pd

# Step 1: Load the built-in dataset
data = load_breast_cancer()

# Step 2: Create DataFrame
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)

# Step 3: Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Step 4: Create Logistic Regression model
model = LogisticRegression(max_iter=10000)

# Step 5: Train the model
model.fit(X_train, y_train)

# Step 6: Make predictions
y_pred = model.predict(X_test)

# Step 7: Print Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Step 8: Print Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Step 9: Print Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Step 10: Compare Actual vs Predicted
comparison = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print("\nActual vs Predicted:")
print(comparison.head(10))

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Accuracy: 0.956140350877193
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Confusion Matrix:
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[[39  4]
 [ 1 70]]
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Classification Report:
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	precision	recall	f1-score	support
0	0.97	0.91	0.94	43
1	0.95	0.99	0.97	71
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

```
Actual vs Predicted:
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